AEP as the Agentic-Container OS

High-Level Design (HLD) · v1.1 · Status: Proposed · Branch: `docs/dual-lane-governance-proposal` · Companion: `aep-lld-v1.1.html`
Supersedes: v1 (dual-lane governance proposal, `docs/draft-proposal.html` v1)

1. Purpose and scope

This document consolidates the design decisions reached in the working discussion into one reference: how the Autonomous Engineering Platform (AEP) is built and deployed portably, and how it governs two kinds of autonomous execution — fast local **ralph-loop** lanes and sandboxed **ESF** (Engineering Software Factory) pipelines — on a shared set of core services.

It is a design, not a record of finished work. Section 9 states exactly what exists in the repository today and what is still to be built; reading the rest of the document as "already implemented" would be wrong.

2. Decision log

#	Decision	Rationale	Status
D1	Orchestration tier is a single Go binary, `aep`; NeuroStrata stays in Rust.	Static cross-compiled binaries need no Python on Windows/embedded hosts; `golang.org/x/sys` gives Job Objects, `flock` / `LockFileEx` and process groups. Python's one real edge (`ast`-based test classification in `mechanical_verify.py`) is narrow.	BUILT (CORE)
D2	Artifact targets: `dist/bin/aep-{linux,darwin}-{amd64,arm64}`, `dist/vm/linux-{amd64,arm64}`, plus OCI image/context.	Covers Linux servers, the 8 GB Muse ARM64 VM, Apple Silicon and Intel Macs, and any Docker/Podman host. Native Windows is explicitly out of the target list (WSL2 or container only).	BUILT
D3	Two execution lanes: Tier 1 ralph-loop, Tier 2 ESF.	Local worktree lanes are fast but share the host kernel; ESF adds microVM isolation and durable workflows for untrusted or high-blast-radius work.	PLANNED
D4	AEP is the "container OS": five shared core services on the substrate; lanes are ephemeral containers above it.	One memory, one ledger, one rate limiter, one policy for every lane — no per-lane drift.	PLANNED
D5	Runtime is **host-family dependent**: on macOS, *one large shared Colima VM* (`vz` + `virtiofs`, sized 12 CPU / 48 GB / 220 GB) with Docker/Podman containers spawned on demand inside it; on Linux, Podman/Docker directly on host. Never one VM per lane in either family.	Container spawning is need-based and admission-bounded inside the single VM/host; per-lane VMs only where kernel-level (microVM) isolation is required, i.e. Tier 2 on `linux-native` with KVM.	PLANNED
D6	`beads` is the single macro task ledger. The Superpowers SDD ledger (`.superpowers/sdd/<plan>/progress.md`) stays a per-plan micro ledger and does not replace it.	Different lifetimes and purposes: durable cross-session backlog vs. ephemeral compaction-recovery record.	PLANNED
D7	ESF workflow IDs are keyed to beads issue IDs (`esf-beads-<ID>`).	Prevents orphan factory runs and dual-backlog drift.	PLANNED
D8	Factory-tagged tasks degrade explicitly when Temporal/CubeSandbox are unreachable (quarantine, or fenced local lane with escalated review) — never hang.	Removes the load-bearing failure found in review: the loop stalling on infrastructure that does not exist on most hosts.	PLANNED

D9	Local-first pluggable Tier-2 sandbox fabric: CubeSandbox (Linux hosts with `/dev/kvm`), `libkrun` (macOS native Hypervisor/Virtualization.framework), and `gVisor` (`runsc` in `systrap` mode for KVM-free Linux/cloud). Remote sandbox clusters are strictly an emergency fallback of last resort if all local sandbox runtimes fail.	Breaks the hard dependency on Linux nested KVM: macOS runs native microVMs via `libkrun`, while KVM-free edge/cloud nodes isolate untrusted tasks via `gVisor` user-space kernel. **Update 2026-11-09:** nested KVM inside Colima is also *confirmed* (`--nested-virtualization` exposes `/dev/kvm` on M3 Max / Colima 0.10.3), so CubeSandbox-in-Colima is a valid macOS Tier-2 route alongside native `libkrun`.	PLANNED

3. Goals and non-goals

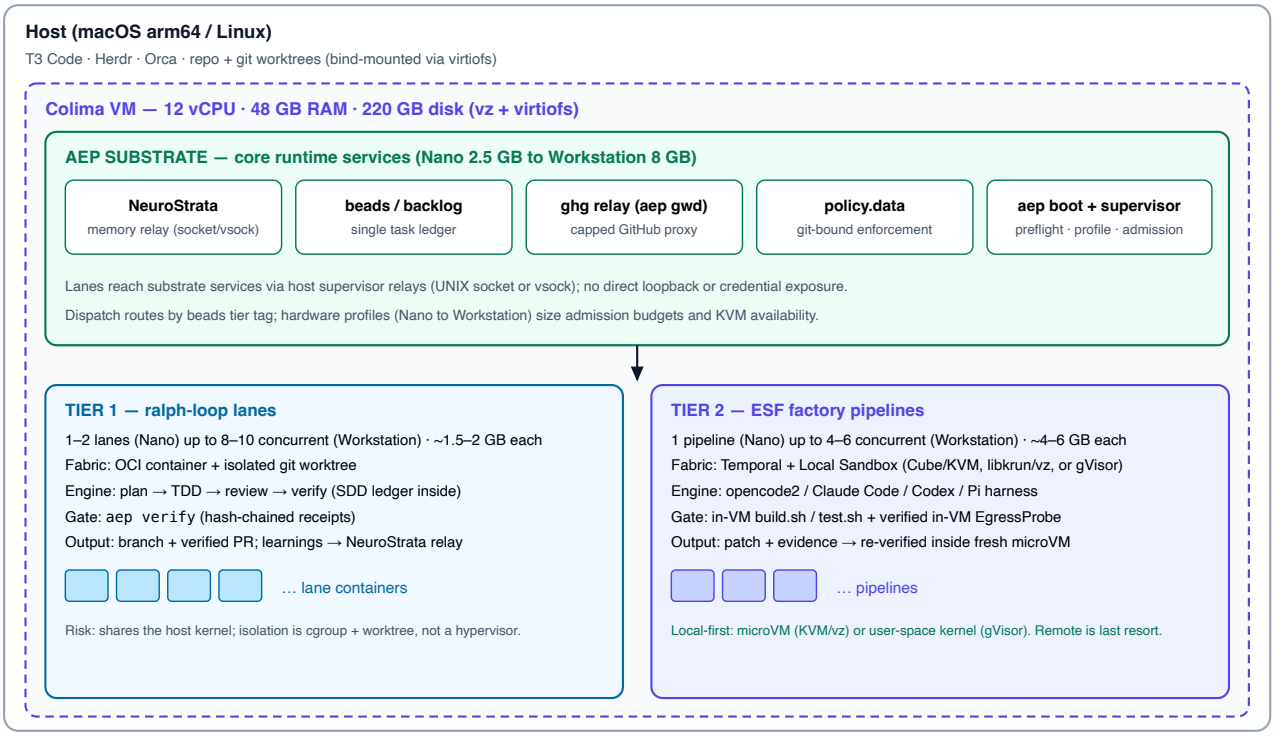
Goals

- One-command install and run on Linux (x86_64 and ARM64), the Muse 8 GB VM, and macOS (Intel and Apple Silicon).
- Many parallel lanes under one governance plane with bounded, predictable resource use.
- Deterministic verification decides success; an agent saying "done" never does.
- Pluggable sandbox isolation for Tier 2: hardware microVMs via CubeSandbox/libkrun, or user-space kernel sandboxing via gVisor on KVM-free hosts.
- Fail closed: no autonomous run starts if the environment is unfit.

Non-goals (v1.1)

- Native Windows binaries or service wrappers.
- macOS guest VM images (macOS runs the native binary under `launchd`).
- Replacing `beads` or NeuroStrata.
- Modifying the vendored `reference/esf` tree (it changes only by re-vendoring).

4. System architecture



5. Components

5.1 Core runtime services (substrate)

Service	Responsibility	Key constraint
NeuroStrata	Hybrid vector + full-text memory (`ladybug.db` , nomic-embed-text-v1.5). Inscription and search via host-mediated socket relay (`aep memoryd`) or vsock outbox.	Loopback `127.0.0.1:34343` private to host; lanes connect via UNIX domain socket; embedding cache pre-seeded; `HF_HUB_OFFLINE=1` ; DB backed up by change-detected watchdog.
beads / backlog	Single canonical task ledger; carries the tier tag and the foreign key used by ESF workflow IDs.	Telemetry off (`bd metrics off`); Dolt remote host allowlisted if egress is blocked.
ghg	Counted, coalescing GitHub gateway: served to lanes via `aep gwd` IPC relay; 3,500/5,000 core req/h self-cap, one poller per watched run.	Lanes communicate via client shim to relay socket; GitHub PAT credentials remain strictly on host.
policy.data	Env-variable scrub fragments, sensitive-path patterns, hardware profile; bound to the committed git blob.	Edits land via PR; boot refuses a policy that differs from `HEAD` .
aep boot / supervisor	Fail-closed preflight, dispatch, capacity admission, profile auto-detection.	Exit non-zero rather than start in a degraded state.

5.2 Execution lanes

	Tier 1 · ralph-loop	Tier 2 · ESF
Intended work	Features, fixes, refactors, tests, docs	Untrusted dependencies, CVE remediation, multi-repo or high-blast-radius change
Isolation	Container + git worktree + process group / Job Object	Pluggable local-first sandbox: Hardware microVM (CubeSandbox on Linux KVM, libkrun on macOS vz) OR User-space kernel (gVisor runsc systrap on KVM-free hosts) per run, destroyed on exit. Remote cluster is strictly an emergency last resort.
Orchestration	Scheduler tick (default 2 h), ≤6 iterations, 3-strike stop	Temporal durable workflow with replay-safe activities
Verification	`aep verify`	Repo `build.sh` / `test.sh` in the microVM (target: bake `aep verify` into the template)
Human gate	PR review; circuit breaker	`factory review --approve/--reject`

6. Scalable resource model (Nano 2vCPU / 8GB to Workstation 12vCPU / 48GB)

Profiles are two-axis — **Host Family** × **Size Tier**. Host families select the local sandbox engine: `linux-kvm` (CubeSandbox microVM via `/dev/kvm`), `mac-native-vz` (hardware microVM via `libkrun` / Apple `Virtualization.framework`), or `linux-gvisor` (user-space kernel via `runsc systrap` without KVM). Size tiers set admission budgets:

Profile	Target hardware	Allocation	Substrate	Tier 1 (Ralph)	Tier 2 (ESF Local Backend)	Isolation Mechanism
`nano-8g`	Intel N100, RPi 5, Cloud 2vCPU/8GB	2 vCPU · 8 GB · 30 GB	1 vCPU · 2.5 GB · 10 GB	4.0 GB budget (1–2 lanes @ 2GB)	1.5 GB budget (1 lane via gVisor systrap)	User-space kernel (no KVM needed)
`small-16g`	Intel Core i5, AMD Ryzen 5, Cloud 4vCPU/16GB	4 vCPU · 16 GB · 80 GB	1 vCPU · 3.5 GB · 15 GB	6.5 GB budget (2–3 lanes @ 2GB)	6.0 GB budget (1 pipeline @ 6GB)	Cube/KVM (Linux) or gVisor (Cloud)
`medium-32g`	Intel Core i7/i9, AMD Ryzen 7/9, Cloud 8vCPU/32GB	8 vCPU · 32 GB · 150 GB	2 vCPU · 4.5 GB · 25 GB	12.0 GB budget (4–6 lanes @ 2GB)	15.0 GB budget (2–3 pipelines @ 5GB)	Cube/KVM (Linux) or libkrun (macOS)
`workstation-48g`	Apple Silicon M3 Max / Linux Threadripper	12 vCPU · 48 GB · 220 GB	2 vCPU · 8.0 GB · 30 GB	16.0 GB budget (8–10 lanes @ 1.5–2GB)	24.0 GB budget (4–6 pipelines @ 4–6GB)	libkrun/vz (macOS) or Cube/KVM (Linux)

Nested-virtualization evidence (2026-11-09).
Confirmed 2026-11-09 on Apple M3 Max (macOS 27.2, Colima 0.10.3, `vz` + `virtiofs`): `colima start --nested-virtualization` exposes `/dev/kvm` inside the VM (and in `--privileged` containers); QEMU `accel=kvm` is accepted for a nested L2 guest; Docker 29.5.2 containers run in-VM; Temporal (sqlite dev server) serves UI on `:8233`. Without `--nested-virtualization`, `/dev/kvm` is absent — the flag is required. CubeSandbox itself was not installed/tested.

Local-first isolation hierarchy.
AEP prioritizes on-node isolation: (1) native hardware microVMs via CubeSandbox/KVM on Linux or `libkrun` / `vz` on macOS; (2) user-space kernel isolation via `gVisor` (`runsc` in `systrap` mode) when hardware virtualization is absent (e.g. Intel N100 or non-metal Cloud VMs). Remote sandbox clusters are strictly an emergency fallback of last resort if all local sandbox runtimes fail.

7. Platform and artifact coverage

Artifact	Covers	Does not cover
`dist/bin/aep-linux-amd64`	Linux x86_64 servers, cloud VMs, CI runners	32-bit x86; native Windows
`dist/bin/aep-linux-arm64`	Muse ARM64 VM, Graviton/Ampere, Raspberry Pi 4/5 (64-bit)	32-bit ARM
`dist/bin/aep-darwin-arm64`	Apple Silicon Macs	Intel Macs (Rosetta only)
`dist/bin/aep-darwin-amd64`	Intel Macs	—
`dist/vm/linux-{amd64,arm64}`	Cloud-init + systemd bundles with per-arch hardware profile	Non-systemd distros
OCI image / `dist/oci` context	Any Docker/Podman host; Kubernetes-style schedulers	Hosts with no container runtime

8. Dispatch and lifecycle

1. **Intake.** Work is a beads issue carrying a tier tag (`local` or `factory`) and acceptance criteria.
2. **Preflight.** `aep boot` must pass: binaries, NeuroStrata answering, standing guards present, policy blob matches `HEAD` , memory floor and hardware profile met.
3. **Admission.** Dispatcher reserves RAM/CPU for the lane against the active profile budget; if it cannot, the task waits (bounded) rather than oversubscribing.
4. **Execute.** Tier 1: claim worktree, start lane container with mounted supervisor socket relays (`aep memoryd` , `aep gwd`), run plan/TDD/review/verify. Tier 2: submit `factory run` with workflow ID `esf-beads-<ID>` .
5. **Verify.** Both lanes emit SHA-256 receipts. For Tier-2 factory work containing untrusted code, independent patch re-verification runs inside a fresh, disposable Tier-2 microVM sandbox with closed egress—never directly on the shared Tier-1 host kernel.
6. **Close.** PR merges, `bd close` , learnings inscribed to NeuroStrata via supervisor relay, lane and reservation released.

9. Governance invariants

- No lane starts unless `aep boot` passes (fail closed).
- All GitHub traffic from lanes goes through `ghg` via the `aep gwd` IPC relay. Lane containers hold no GitHub credentials and cannot connect directly to api.github.com.
- `policy.data` must equal the committed blob; lanes run with credential-named env vars scrubbed.
- ESF runs require `hardening.acknowledge_open_egress = false` (the upstream default), configured closed egress policies (`allow_internet = false`), and verified in-VM `EgressProbe` digest. Checking only the acknowledgement boolean is not network enforcement.

- Tasks requiring Tier-2 isolation MUST NOT be downgraded silently to Tier 1 when Tier 2 / KVM is unavailable; they are strictly quarantined. Downgrade requires an explicit, authenticated operator risk reassessment override (`aep dispatch --override-tier1-risk-reassessment`).
- Independent re-verification of Tier-2 patches must execute inside a fresh hardware-isolated microVM sandbox. Tier 1 performs static analysis only (syntax, diff, lint).
- Model tiering: ralph lanes at `standard` / `measured` ; `frontier` / `thorough` reserved for factory runs.
- A task has exactly one ledger identity (the beads ID) across both lanes.

10. Implementation status (as of this document)

BUILT in `develop`

- `cmd/aep` with `version` , `info` , `boot` , `verify` , `supervisor` , `lock-test` .
- `internal/platform/{memory,lock,process}` for Linux, macOS and Windows; `internal/verifier/runner` (allowlist discovery, tail-capped output, SHA-256 receipts, memory preflight, budgets).
- `Makefile` targets `binaries` , `vm` , `oci` ; `scripts/package-vm.sh` , `scripts/package-oci.sh` ; `Containerfile` , `docker-compose.yml` .
- Root CI runs `gofmt` , `go vet` , `go test` .

NOT YET BUILT

- `aep boot` today checks only the memory floor and platform; the NeuroStrata/guard/gh-auth/policy checks still live in `scripts/boot-gate.sh` .
- `aep supervisor` today runs boot gate + one verification pass per tick; it does not dispatch lanes or ESF runs.
- Lane coordination remains in `scripts/lane-coord.sh` (Linux `flock`); no Go lane manager, no admission control, no beads adapter, no dispatcher, no ESF client, no Colima profile or compose topology for the substrate.
- The "unified verifier inside the CubeSandbox template" and NeuroStrata hooks in the ESF harness are design intent only.

11. Risks and open questions

Risk / question	Why it matters	Cheapest check
MED KVM inside Colima/VZ requires `--nested-virtualization` — **confirmed 2026-11-09**; CubeSandbox itself still unvalidated	CubeSandbox microVMs require Linux KVM. On macOS, KVM exists inside the Colima VM *only* with `--nested-virtualization` (confirmed: `/dev/kvm` present, QEMU `accel=kvm` accepted on M3 Max / Colima 0.10.3). Additional local-first backends remain: `libkrun` on macOS (native `vz`) and `gVisor` (`runsc systrap` on KVM-free Linux). Remote cluster is strictly an emergency last resort.	`colima ssh -- ls -l /dev/kvm` in the VM; probe at `aep boot` and set `Tier2Supported` from the result, never from host family alone. Next: validate CubeSandbox-in-Colima.
HIGH CubeSandbox and Temporal are external	ESF discovers an existing CubeSandbox; it never installs one. Neither exists on this host.	`factory doctor` against the target environment.
MED Lane footprints are unmeasured	Admission thresholds depend on real RSS of ralph and ESF lanes across hardware profiles.	Install Colima/Podman, run representative lanes, record peak RSS.
MED Tier-1 isolation is container-only	Lane containers share one kernel; a hostile task in Tier 1 is not contained like Tier 2.	Keep untrusted work routed to Tier 2; enforce quarantine on fallback.
MED Boot parity gap	Moving to `aep boot` without the daemon/guard/policy checks would silently weaken governance versus `boot-gate.sh`.	Port checks with tests before retiring the shell gate.
LOW Singular/plural docs path	This document lives in `docs/plan/`; the repo already has `docs/plans/` and `docs/superpowers/plans/`.	Decide on one location.

12. Review status & PR #133 review remediations

The v1.1 design incorporates remediations from PR #133 review comments (C1–C5) evaluated under the `receiving-code-review`, `paperthin`, `overseer`, and `factchk` skills: (1) corrected ESF egress acknowledgement vs probe-verified network boundary; (2) strictly prohibited automatic downgrade to Tier 1, enforcing quarantine for isolation-required tasks; (3) specified supervisor socket relays (`aep memoryd` and `aep gwd`) overcoming container loopback and ghg CLI limitations; (4) restricted patch re-verification of untrusted code to disposable microVMs; and (5) formulated a scalable 4-tier profile matrix spanning 2 vCPU / 8 GB (Nano) to 12 vCPU / 48 GB (Workstation).
